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Reconfigurable and scalable manufacturing equipment can be quickly assembled to respond to fast-changing consumer preferences. Intelligent machines can monitor their own health, throughput and yield, and report corrective action to the maintenance staff

installation of these electronic workers in a wide range of industries. The demand for robots to replace highly skilled labour is expected to increase manifold over time.

Issues not to be ignored

There has been a lot of progress made to handle machine failures by providing various levels of redundancy to the control system. Redundancy provides the equipment with an alternate lifeline in case of any failure. Proper handling of the annunciated failures ensures least risk to the operator and minimum downtime.

However, with systems becoming more open and Web-enabled, these have also become more prone to external attacks. What is important to understand by designers and users is that security is a process and not a product. Hence designers have to maintain confidentiality and integrity of the systems, whereas levels of authentication and electronic audits need to be implemented and practised by end users. Firewalls and antivirus would do their bit of preventing invasions.

Coming soon

A no-man factory will be a reality sooner than any of us can imagine. Intelligent machines that communicate

with each other and robots performing tasks to be done by humans will replace labour in many processes. High-end enterprise resource planning applications that integrate the different business processes would help to automate the entire decision-making process with humans expected to define and set the initial business rules of operation. The key factor which will influence the timelines will be the trade-off between the use of expensive technology and availability of labour.

In fact, the concept of fully automated factories has already existed for some time now: customers order online, with electronic transactions that negotiate batch size (in some cases as small as one), price, size and colour; intelligent robots and sophisticated machines smoothly and rapidly fabricate a variety of customised products on demand.

However, as automated factories and processes are too expensive to be rebuilt for every modification and design change, these have to be highly configurable and flexible. Successfully reconfiguring an entire production line or process requires direct access to most of its control elements—switches, valves, motors and drives—down to a fine level of detail.

Remote-controlled automation is finally making headway in manufacturing settings and maintenance applications. The decades-old machine-based vision of automation—powerful super-robots without people to tend them—underestimated the importance of communications. But today, this is purely a matter of networked intelligence which is now well developed and widely available. ●

Major contributors to this report

1. Arvind Kakru, India marketing leader, GE Fanuc
2. Manish Walia, country business head, industrial automation, Delta Electronics India
3. Sriram Moorthy, head-delivery and analytics, ConnectM Technology Solutions
4. Rockwell Automation

The author is a deputy editor at EFY